

SIMATS ENGINEERING

CSA1511 – CLOUD COMPUTING AND BIG DATA ANALYTICS

Name : NIVASINI VM

Reg No : 192511356

Faculty : Dr.HEMAVATHI R

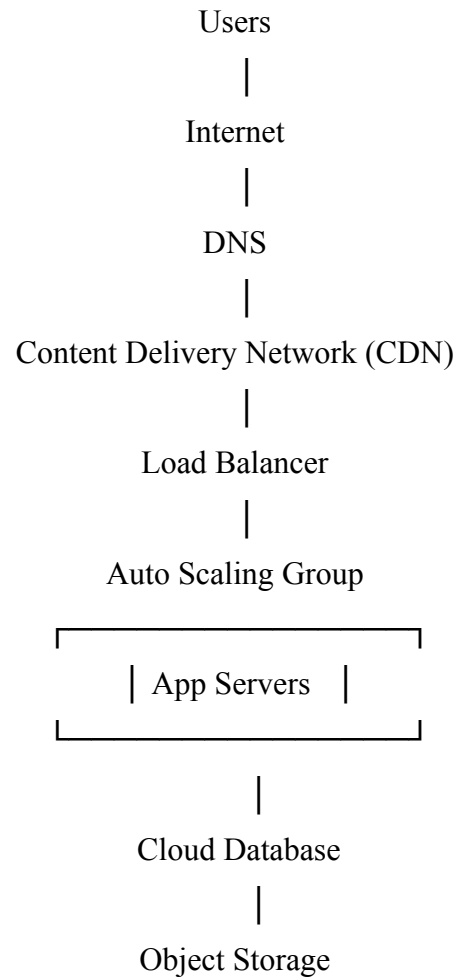
1.A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Scenario Understanding

A media company is migrating its video streaming platform to the cloud to improve scalability, performance, and reliability. The platform must support millions of users who may watch videos simultaneously, especially during live events or popular content releases. Traditional infrastructure cannot easily handle sudden traffic spikes, making cloud computing the ideal solution. The cloud provides flexible resources that automatically increase or decrease according to demand, ensuring uninterrupted streaming and a better user experience.

Cloud Architecture Design

The proposed cloud architecture consists of the following components:



Architecture Explanation:

- Users access the streaming application through the Internet.
- DNS directs users to the nearest Content Delivery Network (CDN).
- The CDN stores frequently watched videos at edge locations, reducing latency and buffering.
- If the requested content is unavailable in the CDN, the request reaches the Load Balancer.
- The Load Balancer distributes traffic equally among multiple application servers.
- Auto Scaling automatically adds or removes virtual machines depending on the number of users.
- Video files are stored in cloud object storage, while metadata is maintained in the cloud database.

Scalability and Caching Techniques

Scalability enables the system to handle increasing workloads without reducing performance. Horizontal scaling is preferred because additional virtual machines can be added automatically during heavy traffic.

Caching improves performance by storing frequently accessed content closer to users. A Content Delivery Network (CDN) is the most effective caching technique for video streaming because it serves videos from edge servers located near users. Application caching and database caching also reduce processing time by storing frequently accessed information in memory.

These techniques reduce server load, improve response time, minimize buffering, and enhance the overall user experience.

Load Balancing and Auto Scaling

Load balancing distributes incoming requests across multiple servers instead of allowing a single server to handle all requests. This prevents server overload and ensures that resources are efficiently utilized. If one server becomes unavailable, the Load Balancer redirects traffic to healthy servers, maintaining uninterrupted service.

Auto Scaling continuously monitors system performance. When user traffic increases, additional virtual machines are launched automatically. When traffic decreases, unnecessary servers are removed. This ensures that sufficient computing resources are always available while avoiding unnecessary operational costs.

Together, Load Balancing and Auto Scaling provide high availability, fault tolerance, and smooth streaming even during peak traffic conditions.

Cost Optimization Strategies

Cloud services can become expensive if resources are not managed efficiently. The following strategies help reduce operational costs:

- Use Auto Scaling to run servers only when required.
- Store rarely accessed videos in low-cost archive storage.
- Compress video files to reduce storage and bandwidth usage.
- Use Reserved Instances for predictable workloads.
- Deliver content through CDN to reduce bandwidth costs.
- Monitor cloud resources regularly and remove unused virtual machines or storage.

These practices help the organization maintain excellent performance while minimizing cloud expenses.

Benefits of the Proposed Solution

The proposed architecture provides several advantages:

- Supports millions of concurrent users.
- Reduces video buffering through CDN caching.
- Automatically adjusts resources based on workload.
- Provides high availability and fault tolerance.
- Improves application performance.

Conclusion

Migrating the video streaming platform to the cloud enables the media company to deliver reliable and high-performance services to millions of users. The combination of CDN, Load Balancer, Auto Scaling, Cloud Storage, and Database services ensures fast content delivery, reduced buffering, and continuous availability. Cost optimization strategies such as Auto Scaling, archive storage, and resource monitoring further reduce operational expenses. Therefore, the proposed cloud architecture is scalable, cost-effective, and capable of meeting the demands of modern video streaming applications.

2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

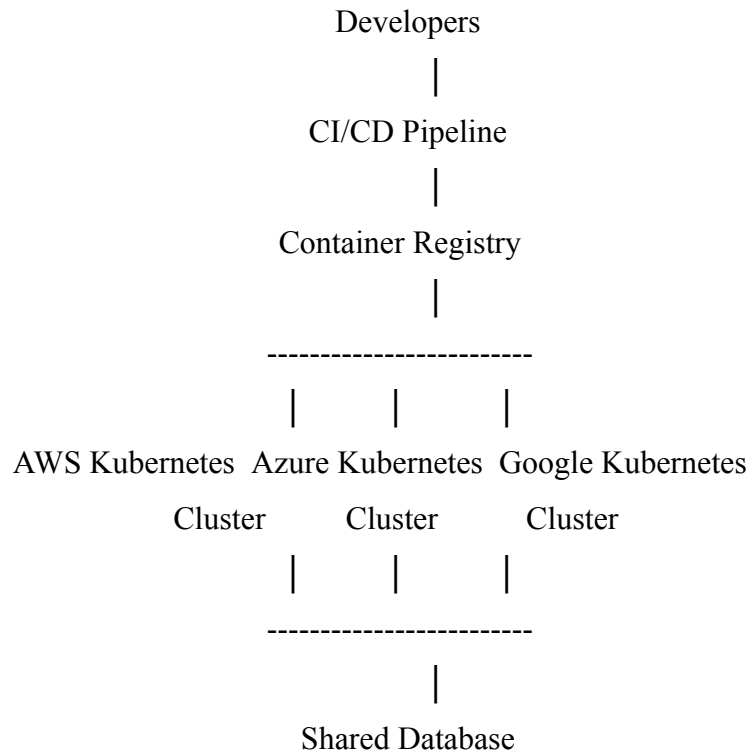
Scenario Understanding

A fintech startup is deploying its applications across multiple cloud providers to improve availability, scalability, and disaster recovery. Since financial applications require high reliability and security, the organization must ensure that the application behaves consistently across all cloud environments. Managing containers manually in different cloud platforms is difficult and may lead to deployment errors, inconsistent configurations, and increased operational complexity. Therefore, a container orchestration platform and a multi-cloud strategy are required to simplify deployment and management.

Proposed Solution

The startup should use **Docker** to package applications into containers and **Kubernetes** as the container orchestration platform. Kubernetes provides automated deployment, scaling, monitoring, and management of containers across multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud Platform.

Proposed Architecture



Architecture Explanation

- Developers build the application and push the container image to a Container Registry.
- The CI/CD pipeline automatically deploys the application to Kubernetes clusters running on different cloud providers.
- Kubernetes manages all containers and ensures consistent deployment.
- A shared database stores application data securely.
- If one cloud provider experiences a failure, traffic can be redirected to another cloud without affecting users.

Benefits of Multi-Cloud Deployment

Using multiple cloud providers offers several advantages:

- **High Availability:** Applications remain accessible even if one cloud provider experiences an outage.
- **Disaster Recovery:** Data and applications can be recovered quickly from another cloud environment.

- **Reduced Vendor Lock-in:** Organizations are not dependent on a single cloud provider.
- **Better Performance:** Users can access services from the nearest cloud region, reducing latency.

Risks of Multi-Cloud Deployment

Although multi-cloud provides many benefits, it also introduces certain challenges:

- Increased management complexity due to multiple cloud platforms.
- Different security policies and configurations across providers.
- Higher networking and data transfer costs.
- Difficulty in maintaining data consistency between clouds.
- More complex monitoring and troubleshooting.
- Need for skilled administrators to manage different cloud environments.

Proper planning, automation, and centralized monitoring tools help reduce these challenges.

Decision Justification

Using Docker and Kubernetes provides a standardized deployment environment, ensuring that applications run consistently across all cloud providers. Kubernetes automates deployment, monitoring, scaling, and recovery, reducing manual effort and improving reliability. A multi-cloud strategy increases availability, minimizes downtime, and protects the organization from vendor lock-in. Service discovery and auto-scaling improve communication between application components and ensure that resources are used efficiently during varying workloads.

Conclusion

A combination of Docker, Kubernetes, and a multi-cloud deployment strategy provides a reliable, scalable, and secure solution for the fintech startup. Kubernetes simplifies container management through automated deployment, service discovery, health monitoring, and scaling. Multi-cloud deployment improves availability, disaster recovery, and business continuity while reducing dependence on a single cloud provider.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

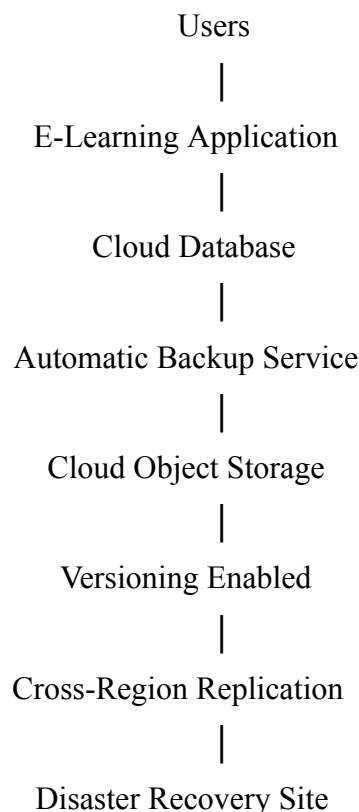
Scenario Understanding

An e-learning platform stores important data such as student records, course materials, assignments, examination results, and user accounts in the cloud.. To prevent such incidents in the future, the organization requires a reliable backup and disaster recovery (DR) solution.

Proposed Cloud Backup and Disaster Recovery Solution

The organization should implement a cloud-native backup strategy that includes automatic backups, data replication, object versioning, and disaster recovery services.

Proposed Architecture



Architecture Explanation

- The application stores all student and course information in a cloud database.
- Automatic backup services create scheduled backups of the database and storage.
- Object versioning stores previous versions of files, allowing deleted or modified files to be restored.

Cloud-Native Backup, Replication, and Versioning

Automatic Backup

Cloud backup services create regular backups without manual intervention. Daily and incremental backups reduce the risk of permanent data loss and allow quick restoration of important information.

Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

Recovery Time Objective (RTO)

Recovery Time Objective is the maximum acceptable time required to restore services after a failure.

For the e-learning platform, an RTO of **30 minutes** can be achieved by maintaining standby resources and automated disaster recovery systems. This ensures that students and teachers experience minimal downtime.

Business Continuity During Failures

To ensure uninterrupted services, the organization should follow these steps:

- Enable automatic daily and incremental backups.
- Configure object versioning for all important files.
- Replicate data across multiple cloud regions.
- Regularly test backup restoration procedures.
- Monitor cloud resources continuously using cloud monitoring tools.

Benefits of the Proposed Solution

The proposed disaster recovery strategy provides several advantages:

- Protects against accidental deletion and data corruption.
- Enables quick restoration of lost files.
- Reduces downtime during failures.
- Improves data availability and reliability.
- Supports business continuity.
- Enhances student and faculty confidence in the system.
- Ensures secure storage of educational data.

Decision Justification

The combination of automatic backups, cross-region replication, and object versioning provides comprehensive protection against data loss. Regular backups ensure data can be restored, while replication protects against regional failures. Versioning allows recovery of deleted or modified files without affecting other data. Achieving low RTO and RPO values ensures fast recovery with minimal data loss, making the proposed solution reliable and suitable for an e-learning environment.

Conclusion

A cloud-native backup and disaster recovery strategy is essential for protecting the e-learning platform from accidental deletion and unexpected failures. Automatic backups, replication, and versioning work together to ensure data availability, quick recovery, and continuous service. By maintaining low RTO and RPO values and implementing effective business continuity practices, the organization can provide a secure, reliable, and uninterrupted learning experience for students and educators.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

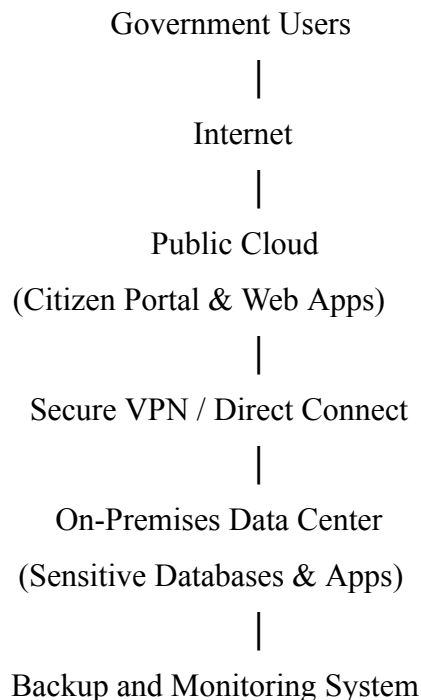
Scenario Understanding

A government agency manages both sensitive and non-sensitive information. Sensitive data such as citizen records, with government regulations. Therefore, the agency requires a **hybrid cloud architecture** that combines the security of on-premises infrastructure with the scalability and flexibility of the public cloud.

Proposed Hybrid Cloud Architecture

The hybrid cloud architecture consists of an on-premises data center connected securely to a public cloud environment.

Architecture Diagram



Architecture Explanation

- Public-facing applications such as citizen portals are deployed in the public cloud.
- Sensitive applications and databases remain within the on-premises data center.
- A secure VPN or dedicated private connection is established between the cloud and the data center.
- Backup, monitoring, and security tools continuously protect both environments.
- Users access services securely without knowing where the applications are hosted.

Identity Management

Identity management controls who can access cloud resources and government applications.

The agency should implement:

- **Single Sign-On (SSO)** to allow users to access multiple systems with one login.
- **Multi-Factor Authentication (MFA)** to provide an additional layer of security.
- **Role-Based Access Control (RBAC)** to assign permissions based on user roles.
- **Identity Federation** to synchronize user identities between the on-premises environment and the cloud.

These techniques improve security while simplifying user authentication and administration

Encryption

Encryption protects sensitive government information from unauthorized access.

Two types of encryption should be implemented:

Challenges in Managing Hybrid Cloud Environments

Although hybrid cloud provides many advantages, it also introduces several management challenges.

- Managing infrastructure across both cloud and on-premises environments.

- Ensuring consistent security policies.
- Maintaining regulatory compliance.
- Integrating different platforms and technologies.
- Monitoring applications across multiple environments.
- Higher operational and maintenance costs.
- Managing network latency between cloud and on-premises systems.

Proper planning, centralized monitoring, and automation tools help overcome these challenges.

Benefits of the Proposed Solution

The proposed hybrid cloud architecture offers several advantages:

- Protects sensitive government data.
- Supports regulatory compliance.
- Provides scalability for public services.
- Enables secure communication between environments.
- Improves system availability and reliability.

Decision Justification

A hybrid cloud architecture is the most suitable solution because it allows the agency to keep confidential information on-premises while using public cloud resources for non-sensitive applications. Secure VPN connections, encryption, identity management, and access control ensure that data remains protected at all times.

Conclusion

The proposed hybrid cloud architecture combines the advantages of private and public cloud environments. Sensitive government data remains securely stored on-premises, while public services benefit from the scalability and flexibility of the cloud. Secure communication, strong identity management, encryption, and continuous monitoring ensure data protection and compliance. Although hybrid cloud management introduces additional complexity, it provides an effective and secure solution for modern government organizations.

5. SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

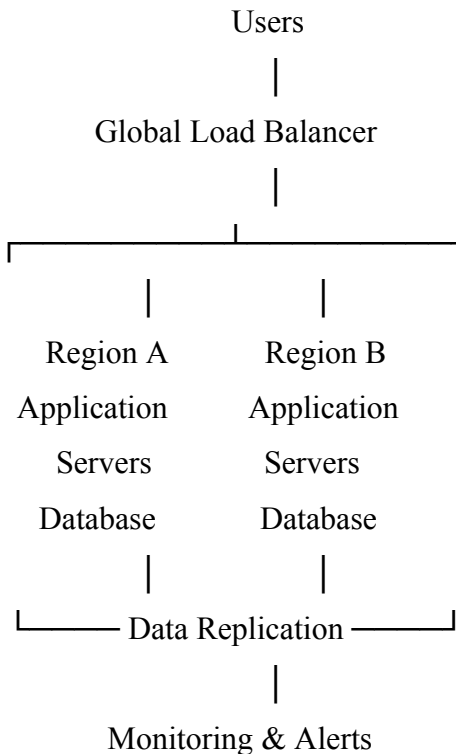
Scenario Understanding

A Software as a Service (SaaS) provider delivers applications to customers over the internet. Since users depend on the service continuously, the application must remain available even if an entire cloud region experiences a failure due to hardware issues, power outages, or natural disasters. To achieve high availability and fault tolerance, the SaaS provider should deploy the application across multiple cloud regions. This ensures that users experience little or no downtime during unexpected failures.

Proposed Multi-Region Deployment Strategy

The SaaS application should be deployed in **two or more cloud regions**. Each region contains identical application servers, databases, and storage resources. A global load balancer distributes user requests to the nearest healthy region. If one region becomes unavailable, traffic is automatically redirected to another region.

Architecture Diagram



Architecture Explanation

- Users access the application through a global load balancer.
- Requests are routed to the nearest available region.
- Each region hosts application servers and a database.
- Data is continuously replicated between regions.
- Monitoring services track the health of all resources and generate alerts when failures occur.

Data Replication

Data replication ensures that copies of application data are available in multiple regions.

Two replication methods can be used:

Synchronous Replication

- Data is written to both regions at the same time.
- Provides maximum data consistency.

- Suitable for critical financial or business transactions.

Asynchronous Replication

- Data is first written to the primary region and later copied to the secondary region.
- Offers better performance with minimal delay.
- Suitable for applications where a slight delay in synchronization is acceptable.

By replicating data across regions, the SaaS provider can recover quickly from failures without significant data loss.

Failover Mechanism

A failover mechanism automatically switches user traffic from the failed region to a healthy region.

The process includes:

- Continuous health checks monitor application availability.
- If Region A becomes unavailable, the global load balancer detects the failure.
- User requests are automatically redirected to Region B.
- The replicated database in Region B becomes the active database.
- Once Region A is restored, synchronization resumes and traffic can be balanced again.

This automatic process minimizes downtime and ensures uninterrupted service.

Monitoring and Alerting Strategies

Continuous monitoring is essential for maintaining application performance and detecting issues before they affect users.

The SaaS provider should monitor:

- CPU and memory utilization.
- Network traffic and bandwidth.

- Application response time.
- Database performance.
- Disk usage.
- Error rates and failed requests.
- Server health and availability.

Alerting mechanisms should notify administrators immediately through:

- Email notifications.
- SMS alerts.
- Mobile app notifications.

Benefits of the Proposed Solution

- High availability of applications.
- Fault tolerance during regional failures.
- Faster disaster recovery.
- Improved application performance.

Decision Justification

Deploying the SaaS application across multiple cloud regions ensures that services remain available even during major outages. Data replication protects important information from loss, while automatic failover redirects users without manual intervention. Continuous monitoring and alerting help identify issues early and improve system reliability.

Conclusion

A multi-region cloud deployment is the most effective solution for ensuring high availability and fault tolerance in a SaaS application. By combining global load balancing, data replication, automatic failover, and continuous monitoring, the provider can deliver reliable services with minimal downtime. This approach improves business continuity, protects customer data, and ensures a seamless user experience even when an entire cloud region becomes unavailable.